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● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Geometry and Trigonometry

05

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ANSWER KEY · SAT QUESTION BANK

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Q1**D****D.** $\frac{119}{6}$

Choice D is correct. The figure shows that angle MRL and angle PRQ are vertical angles. Since vertical angles are congruent, angle MRL and angle PRQ are congruent. It's given that $\overline{LM}$ is parallel to $\overline{PQ}$. The figure also shows that $\overline{LQ}$ intersects $\overline{LM}$ and $\overline{PQ}$. If two parallel segments are intersected by a third segment, alternate interior angles are congruent. Thus, alternate interior angles MLR and PQR are congruent. Since triangles LMR and PQR have two pairs of congruent angles, the triangles are similar. Sides LR and MR in triangle LMR correspond to sides RQ and RP , respectively, in triangle PQR . Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{RQ}{LR} = \frac{RP}{MR}$. It's given that the lengths of $\overline{MR}$, $\overline{LR}$, and $\overline{RP}$ are 6, 7, and 11, respectively. Substituting 6 for MR , 7 for LR , and 11 for RP in the equation $\frac{RQ}{LR} = \frac{RP}{MR}$ yields $\frac{RQ}{7} = \frac{11}{6}$. Multiplying each side of this equation by 7 yields $RQ = \frac{11}{6}7$, or $RQ = \frac{77}{6}$. It's given that $\overline{LQ}$ intersects $\overline{MP}$ at point R , so $LQ = LR + RQ$. Substituting 7 for LR and $\frac{77}{6}$ for RQ in this equation yields $LQ = 7 + \frac{77}{6}$, or $LQ = \frac{119}{6}$. Therefore, the length of $\overline{LQ}$ is $\frac{119}{6}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the length of $\overline{RQ}$, not $\overline{LQ}$.

Choice C is incorrect and may result from conceptual or calculation errors.

Q2**B****B. 27**

Choice B is correct. The surface area of a cube with side length s is equal to $6s^2$. Since the surface area is given as 54 square meters, the equation $54 = 6s^2$ can be used to solve for s . Dividing both sides of the equation by 6 yields $9 = s^2$. Taking the square root of both sides of this equation yields $3 = s$ and $-3 = s$. Since the side length of a cube must be a positive value, $s = -3$ can be discarded as a possible solution, leaving $s = 3$. The volume of a cube with side length s is equal to s^3 . Therefore, the volume of this cube, in cubic meters, is 3^3 , or 27.

Choices A, C, and D are incorrect and may result from calculation errors.

Q3**B****B. 13π**

Choice B is correct. The ratio of the lengths of two arcs of a circle is equal to the ratio of the measures of the central angles that subtend the arcs. It's given that arc $\overline{ADC}$ is subtended by a central angle with measure 100° . Since the sum of the measures of the angles about a point is 360° , it follows that arc $\overline{ABC}$ is subtended by a central angle with measure

$360^\circ - 100^\circ = 260^\circ$. If s is the length of arc $\overline{ABC}$, then s must satisfy the ratio $\frac{s}{5\pi} = \frac{260}{100}$.

Reducing the fraction $\frac{260}{100}$ to its simplest form gives $\frac{13}{5}$. Therefore, $\frac{s}{5\pi} = \frac{13}{5}$. Multiplying both sides of $\frac{s}{5\pi} = \frac{13}{5}$ by 5π yields $s = 13\pi$.

Choice A is incorrect. This is the length of an arc consisting of exactly half of the circle, but arc $\overline{ABC}$ is greater than half of the circle. Choice C is incorrect. This is the total circumference of the circle. Choice D is incorrect. This is half the length of arc $\overline{ABC}$, not its full length.

Q4**B****B.** $47\sqrt{2}$

Choice B is correct. It's given that the right triangle is isosceles. In an isosceles right triangle, the two legs have equal lengths, and the length of the hypotenuse is $\sqrt{2}$ times the length of one of the legs. Let l represent the length, in inches, of each leg of the isosceles right triangle. It follows that the length of the hypotenuse is $l\sqrt{2}$ inches. The perimeter of a figure is the sum of the lengths of the sides of the figure. Therefore, the perimeter of the isosceles right triangle is $l + l + l\sqrt{2}$ inches. It's given that the perimeter of the triangle is $94 + 94\sqrt{2}$ inches. It follows that $l + l + l\sqrt{2} = 94 + 94\sqrt{2}$. Factoring the left-hand side of this equation yields $1 + 1 + \sqrt{2} l = 94 + 94\sqrt{2}$, or $2 + \sqrt{2} l = 94 + 94\sqrt{2}$. Dividing both sides of this equation by $2 + \sqrt{2}$ yields $l = \frac{94 + 94\sqrt{2}}{2 + \sqrt{2}}$. Rationalizing the denominator of the right-hand side of this equation by multiplying the right-hand side of the equation by $\frac{2 - \sqrt{2}}{2 - \sqrt{2}}$ yields $l = \frac{94 + 94\sqrt{2} \cdot 2 - \sqrt{2}}{2 + \sqrt{2} \cdot 2 - \sqrt{2}}$. Applying the distributive property to the numerator and to the denominator of the right-hand side of this equation yields $l = \frac{188 + 94\sqrt{2} + 188\sqrt{2} - 94\sqrt{4}}{4 + 2\sqrt{2} + 2\sqrt{2} - \sqrt{4}}$. This is equivalent to $l = \frac{94\sqrt{2}}{2}$, or $l = 47\sqrt{2}$. Therefore, the length, in inches, of one leg of the isosceles right triangle is $47\sqrt{2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the length, in inches, of the hypotenuse.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5

14.66

Accept also: 14.67, 44/3

The correct answer is $\frac{44}{3}$. It's given that in triangle RST , angle T is a right angle. The area of a right triangle can be found using the formula $A = \frac{1}{2}l_1l_2$, where A represents the area of the right triangle, l_1 represents the length of one leg of the triangle, and l_2 represents the length of the other leg of the triangle. In triangle RST , the two legs are $\vec{RT}$ and $\vec{ST}$. Therefore, if the length of $\vec{RT}$ is 72 and the area of triangle RST is 792, then $792 = \frac{1}{2}72ST$, or $792 = 36ST$. Dividing both sides of this equation by 36 yields $22 = ST$. Therefore, the length of $\vec{ST}$ is 22. It's also given that point L lies on $\vec{RS}$, point K lies on $\vec{ST}$, and $\vec{LK}$ is parallel to $\vec{RT}$. It follows that angle LKS is a right angle. Since triangles RST and LSK share angle S and have right angles T and K , respectively, triangles RST and LSK are similar triangles. Therefore, the ratio of the length of $\vec{RT}$ to the length of $\vec{LK}$ is equal to the ratio of the length of $\vec{ST}$ to the length of $\vec{SK}$. If the length of $\vec{RT}$ is 72 and the length of $\vec{LK}$ is 24, it follows that the ratio of the length of $\vec{RT}$ to the length of $\vec{LK}$ is $\frac{72}{24}$, or 3, so the ratio of the length of $\vec{ST}$ to the length of $\vec{SK}$ is 3. Therefore, $\frac{22}{SK} = 3$. Multiplying both sides of this equation by SK yields $22 = 3SK$. Dividing both sides of this equation by 3 yields $\frac{22}{3} = SK$. Since the length of $\vec{ST}$, 22, is the sum of the length of $\vec{SK}$, $\frac{22}{3}$, and the length of $\vec{KT}$, it follows that the length of $\vec{KT}$ is $22 - \frac{22}{3}$, or $\frac{44}{3}$. Note that 44/3, 14.66, and 14.67 are examples of ways to enter a correct answer.

Q6**B****B. 24**

Choice B is correct. The length of a segment in the xy -plane can be found using the distance formula, $\sqrt{x_2 - x_1^2 + y_2 - y_1^2}$, where x_1, y_1 and x_2, y_2 are the endpoints of the segment. The segment shown has endpoints at $-6, 4$ and $3, 10$. Substituting $-6, 4$ and $3, 10$ for x_1, y_1 and x_2, y_2 , respectively, in the distance formula yields $\sqrt{3 - -6^2 + 10 - 4^2}$, or $\sqrt{9^2 + 6^2}$, which is equivalent to $\sqrt{81 + 36}$, or $\sqrt{117}$. Let x represent the length, in units, of the other leg of this triangle. The area, A , of a right triangle can be calculated using the formula $A = \frac{1}{2}bh$, where b and h are the lengths of the legs of the triangle. It's given that the area of the triangle is $36\sqrt{13}$ square units. Substituting $36\sqrt{13}$ for A , $\sqrt{117}$ for b , and x for h in the formula $A = \frac{1}{2}bh$ yields $36\sqrt{13} = \frac{1}{2}\sqrt{117}x$. Multiplying both sides of this equation by 2 yields $72\sqrt{13} = x\sqrt{117}$. Dividing both sides of this equation by $\sqrt{117}$ yields $\frac{72\sqrt{13}}{\sqrt{117}} = x$. Multiplying the numerator and denominator on the left-hand side of this equation by $\sqrt{117}$ yields $\frac{72\sqrt{1,521}}{117} = x$, or $\frac{7239}{117} = x$, which is equivalent to $\frac{2,808}{117} = x$, or $24 = x$. Therefore, the length, in units, of the other leg of this triangle is 24.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. $3\sqrt{13}$ is equivalent to $\sqrt{117}$, which is the length, in units, of the line segment shown in the xy -plane, not the length, in units, of the other leg of the triangle.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7**A**

A. $x + 2^2 + y + 6^2 = 49$

Choice A is correct. According to the graph, the center of circle A has coordinates $-2, 0$, and the radius of circle A is 3. It's given that circle B is the result of shifting circle A down 6 units and increasing the radius so that the radius of circle B is 2 times the radius of circle A . It follows that the center of circle B is 6 units below the center of circle A . The point that's 6 units below $-2, 0$ has the same x -coordinate as $-2, 0$ and has a y -coordinate that is 6 less than the y -coordinate of $-2, 0$. Therefore, the coordinates of the center of circle B are $-2, 0 - 6$, or $-2, -6$. Since the radius of circle B is 2 times the radius of circle A , the radius of circle B is 6. A circle in the xy -plane can be defined by an equation of the form $(x - h)^2 + (y - k)^2 = r^2$, where the coordinates of the center of the circle are h, k and the radius of the circle is r . Substituting -2 for h , -6 for k , and 6 for r in this equation yields $(x - -2)^2 + (y - -6)^2 = 6^2$, which is equivalent to $(x + 2)^2 + (y + 6)^2 = 36$, or $x^2 + 4x + 2^2 + y^2 + 12y + 6^2 = 36$. Therefore, the equation $x^2 + 4x + 2^2 + y^2 + 12y + 6^2 = 36$ defines circle B .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation defines a circle that's the result of shifting circle A up, not down, by 6 units and increasing the radius.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**.7241**

Accept also: 21/29

The correct answer is $\frac{21}{29}$. It's given that triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angles C and F are right angles. In similar triangles, the tangents of corresponding angles are equal. Therefore, if $\tan A = \frac{21}{20}$, then $\tan D = \frac{21}{20}$. In a right triangle, the tangent of an acute angle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. Therefore, in triangle DEF , if $\tan D = \frac{21}{20}$, the ratio of the length of $\overline{EF}$ to the length of $\overline{DF}$ is $\frac{21}{20}$. If the lengths of $\overline{EF}$ and $\overline{DF}$ are 21 and 20, respectively, then the ratio of the length of $\overline{EF}$ to the length of $\overline{DF}$ is $\frac{21}{20}$. In a right triangle, the sine of an acute angle is the ratio of the length of the leg opposite the angle to the length of the hypotenuse. Therefore, the value of $\sin D$ is the ratio of the length of $\overline{EF}$ to the length of $\overline{DE}$. The length of $\overline{DE}$ can be calculated using the Pythagorean theorem, which states that if the lengths of the legs of a right triangle are a and b and the length of the hypotenuse is c , then $a^2 + b^2 = c^2$. Therefore, if the lengths of $\overline{EF}$ and $\overline{DF}$ are 21 and 20, respectively, then $21^2 + 20^2 = DE^2$, or $841 = DE^2$. Taking the positive square root of both sides of this equation yields $29 = DE$. Therefore, if the lengths of $\overline{EF}$ and $\overline{DF}$ are 21 and 20, respectively, then the length of $\overline{DE}$ is 29 and the ratio of the length of $\overline{EF}$ to the length of $\overline{DE}$ is $\frac{21}{29}$. Thus, if $\tan A = \frac{21}{20}$, the value of $\sin D$ is $\frac{21}{29}$. Note that 21/29, .7241, and 0.724 are examples of ways to enter a correct answer.

Q9**D****D. 135°**

Choice D is correct. Corresponding angles in similar figures have equal measure. It's given that quadrilaterals $PQRS$ and $WXYZ$ are similar and that P , Q , and R correspond to W , X , and Y . It follows that $\angle S$ corresponds to $\angle Z$. It's also given that the measure of $\angle S$ is 135° . Therefore, the measure of $\angle Z$ is 135° .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This is the supplement of the measure of $\angle Z$, not the measure of $\angle Z$.

Q10

C

C. $R = 7r$ and $H = 8h$

Choice C is correct. The volume of a right circular cylinder is equal to $\pi a^2 b$, where a is the radius of a base of the cylinder and b is the height of the cylinder. It's given that the cylinder shown has a radius of r and a height of h . It follows that the volume of the cylinder shown is equal to $\pi r^2 h$. It's given that the second right circular cylinder has a radius of R and a height of H . It follows that the volume of the second cylinder is equal to $\pi R^2 H$. Choice C gives $R = 7r$ and $H = 8h$. Substituting $7r$ for R and $8h$ for H in the expression that represents the volume of the second cylinder yields $\pi 7r^2 8h$, or $\pi 49r^2 8h$, which is equivalent to $\pi 392r^2 h$, or $392\pi r^2 h$. This expression is equal to 392 times the volume of the cylinder shown, $\pi r^2 h$. Therefore, $R = 7r$ and $H = 8h$ could represent the radius R , in terms of r , and the height H , in terms of h , of the second cylinder.

Choice A is incorrect. Substituting $8r$ for R and $7h$ for H in the expression that represents the volume of the second cylinder yields $\pi 8r^2 7h$, or $\pi 64r^2 7h$, which is equivalent to $\pi 448r^2 h$, or $448\pi r^2 h$. This expression is equal to 448, not 392, times the volume of the cylinder shown.

Choice B is incorrect. Substituting $8r$ for R and $49h$ for H in the expression that represents the volume of the second cylinder yields $\pi 8r^2 49h$, or $\pi 64r^2 49h$, which is equivalent to $\pi 3,136r^2 h$, or $3,136\pi r^2 h$. This expression is equal to 3,136, not 392, times the volume of the cylinder shown.

Choice D is incorrect. Substituting $49r$ for R and $8h$ for H in the expression that represents the volume of the second cylinder yields $\pi 49r^2 8h$, or $\pi 2,401r^2 8h$, which is equivalent to $\pi 19,208r^2 h$, or $19,208\pi r^2 h$. This expression is equal to 19,208, not 392, times the volume of the cylinder shown.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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